MAIS PAC

IPhO 04.03: Calculus · Worked Solutions

Calculus: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IPhO questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

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Solution 1 — Rates from a position function. Return →

(a) $v = 6t^2-18t+12$ and $a = 12t-18$.

(b) $6(t^2-3t+2) = 0$ gives $t = 1$ {s} and $t = 2$ {s}.

(c) $x(3) = 54-81+36 = 9$ {m}, and $x(0) = 0$, so the displacement is 9 {m}. Note the particle reverses twice; the **distance** travelled is larger than the displacement. **Check:** a(1) = -6 {m,s^(-2)} and a(2) = +6 {m,s^(-2)}, so the first turning point is a maximum and the second a minimum — exactly what a sketch shows.

Solution 2 — Area under a graph. Return →

(a) A rising straight line to (10, 20), a horizontal line to (30, 20), then a falling line to (35, 0).

(b) Areas: (1)/(2)(10)(20) = 100 {m}, 20×20 = 400 {m}, (1)/(2)(5)(20) = 50 {m}; total 550 {m}.

(c) Average speed = 550/35 = 15.7 {m,s^(-1)}. **Check:** the average must lie between 0 and 20 {m,s^(-1)}, and the long constant-speed phase pulls it close to 20 — the answer is plausible.

Solution 3 — Work from a variable force. Return →

(a) $W = \int_0^x kx',dx' = \frac{1}{2}kx^2$.

(b) $W = \frac{1}{2}(200)(0.10)^2 = 1.0$ {J}.

(c) Work is force times displacement, and on a force–extension graph the area under the line is exactly that product, summed over the stretch. The factor $\frac{1}{2}$ comes because the force starts at zero and rises linearly, so the average force is half the final force. **Check:** the graph area is a triangle, $\frac{1}{2}\times$ base $\times$ height $= \frac{1}{2}(0.10)(20) = 1.0$ {J}.

Solution 4 — A related-rates problem. Return →

(a) $V = \frac{4}{3}\pi r^3$.

(b) $\frac{dV}{dt} = 4\pi r^2\frac{dr}{dt}$.

(c) At r = 10 {cm}: $\frac{dr}{dt} = \frac{50}{4\pi(100)} = 0.040$ {cm,s^(-1)}. At r = 20 {cm}: $\frac{50}{4\pi(400)} = 0.010$ {cm,s^(-1)}. The radius grows four times more slowly once the balloon is twice as big, because the surface area quadruples. **Check:** this is the inverse-square behaviour of a fixed flow spread over an area, and it is why blowing up a balloon gets harder.

Solution 5 — Simple harmonic motion by differentiation. Return →

(a) $v = -A\omega\sin\omega t$ and $a = -A\omega^2\cos\omega t = -\omega^2 x$.

(b) For a spring, $F = -kx = ma$, so $\omega = \sqrt{k/m}$.

(c) v is largest when $\sin\omega t = \pm 1$, that is when $\cos\omega t = 0$ and $x = 0$. **Check:** a is zero there and maximal at the endpoints, which is the phase relationship that makes SHM useful as a model: displacement and acceleration are always exactly antiphase.

Solution 6 — The half-life from an exponential. Return →

(a) $t_{1/2} = \ln 2/k = 0.693/0.0693 = 10.0$ {s}.

(b) After 30 {s} exactly three half-lives have passed, so 1/8 = 0.125 remains.

(c) $A_0 = kN_0 = 0.0693\times 2.0\times 10^{20} = 1.4\times 10^{19}$ {Bq}; after 30 {s} the activity is 1.7×10^{18} {Bq}. **Check:** the activity falls by the same factor, 1/8, as the number of nuclei — activity is proportional to N, which is the definition of first-order decay.

Solution 7 — The binomial approximation. Return →

(a) $\theta_0 = 10^\circ = 0.1745$ {rad}, so $\theta_0^2/16 = 0.0019$: the period is 0.19% longer.

(b) At 30°, $\theta_0 = 0.5236$ {rad} and $\theta_0^2/16 = 0.0171$, a 1.7% correction. The next term is of order θ_0^4, about 0.03%, so the first-order formula is still good to a few parts in 10^4.

(c) Beyond about 45° the first-order term alone underestimates the correction noticeably, and the series converges slowly; at that point the exact elliptic-integral result or a numerical solution is needed. **Check:** at 5° the correction is only 0.05%, which is why a school pendulum is timed to better than a tenth of a percent with a small swing.

Solution 8 — Integrating a force that grows with time. Return →

(a) $J = \int_0^T F_0(1-t/T)\,dt = F_0[t-(t^2)/(2T)]_0^T = (1)/(2)F_0T$.

(b) $v = J/m = (F_0T)/(2m)$.

(c) That is exactly impulse over mass, by the impulse–momentum theorem. Units: F_0T has units {N,s} = {kg,m,s^(-1)}, and dividing by kg gives {m,s^(-1)}. **Check:** the average force is $F_0/2$, which is exactly what the triangular force–time graph shows — a useful shortcut for any linear ramp.

Solution 9 — An RC discharge. Return →

(a) $I = C(dV)/(dt) = -(V_0)/(R)e^{(-t/RC)}$; the sign shows the current flows to discharge the capacitor.

(b) Power $P = I^2R = (V_0^2)/(R)e^{(-2t/RC)}$, so

$$E = \int_0^\infty (V_0^2)/(R)e^{(-2t/RC)}\,dt = (V_0^2)/(R)\cdot(RC)/(2) = (1)/(2)CV_0^2, \qquad (1)$$

which is exactly the stored energy.

(c) The exponential only reaches zero as $t\to\infty$; the charge on the capacitor never vanishes exactly, though it becomes negligible after a few time constants. **Check:** half the stored energy is dissipated in the first 0.35RC, a good quick estimate to remember.

Solution 10 — Terminal speed with a drag force. Return →

(a) m(dv)/(dt) = mg - bv, with solution (from rest)

$$v = (mg)/(b)(1-e^(-bt/m)) = v_({term})(1-e^(-t/\tau)),\ \tau = (m)/(b). \quad (2)$$

(b) Differentiating, a = (dv)/(dt) = ge^(-t/τ), so the acceleration starts at g and decays to zero: the drag grows until it balances the weight.

(c) s = ∫_0^t v,dt' = v_({term})[t - τ(1-e^(-t/τ))]. For t≫τ the bracket tends to t-τ, so s→ v_({term})t - v_({term})τ: constant speed with a constant "lost distance" v_({term})τ. **Check:** setting b = 0 makes τ→∞ and the limit singular, so the drag-free case s = (1)/(2)gt² must be recovered by expanding the exponential instead — a good reminder that a limit can be hidden inside an expansion.